

SHETH L.U.J AND SIR M.V COLLEGE

Artificial Intelligence

Practical No.6

Aim: Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

Code:

```
File Edit Format Run Options Window Help
# Restricted Mode on
C:\Users\gaza\Desktop\pro4.py (3/3/2)
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
df = pd.read_csv('Iris.csv')

# Select input features and target variable
X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Create a weak classifier using a decision stump
weak_model = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

# Train the individual weak classifier
weak_model.fit(X_train, y_train)

# Predict using the weak classifier
plt.bar(models, accuracies)

plt.xlabel("Models")
plt.ylabel("Accuracy (%)")
plt.title("Weak Classifier vs AdaBoost")
plt.ylim(0, 105)

for i, value in enumerate(accuracies):
    plt.text(i, value + 1, f'{value:.2f}%', ha='center')

plt.tight_layout()
plt.show()

# Visualize the AdaBoost confusion matrix
plt.figure(figsize=(6, 5))

plt.imshow(cm)
plt.title("AdaBoost Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.colorbar()

classes = model.classes_

plt.xticks(range(len(classes)), classes, rotation=20)
plt.yticks(range(len(classes)), classes)

# Display values inside the confusion matrix
for i in range(len(classes)):
    for j in range(len(classes)):
        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.tight_layout()
```

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Output:

```
Python 3.13.2 (tags/v3.13.2:4f8bb39, Feb  4 2025, 15:23:48) [MSC v.1942 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

----- RESTART: C:/Users/gazala/OneDrive/Desktop/pno.6.py -----
Weak Classifier Accuracy: 0.6666666666666666
Weak Classifier Accuracy Percentage: 66.66666666666666 %

AdaBoost Accuracy: 0.9333333333333333
AdaBoost Accuracy Percentage: 93.33333333333333 %

AdaBoost Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]

AdaBoost Classification Report:
              precision    recall  f1-score   support

Iris-setosa      1.00        1.00        1.00         10
Iris-versicolor  0.90        0.90        0.90         10
Iris-virginica   0.90        0.90        0.90         10

accuracy          0.93          0.93          0.93         30
macro avg         0.93          0.93          0.93         30
weighted avg      0.93          0.93          0.93         30

Performance Comparison:
Weak Classifier: 66.66666666666666 %
AdaBoost: 93.33333333333333 %
```

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